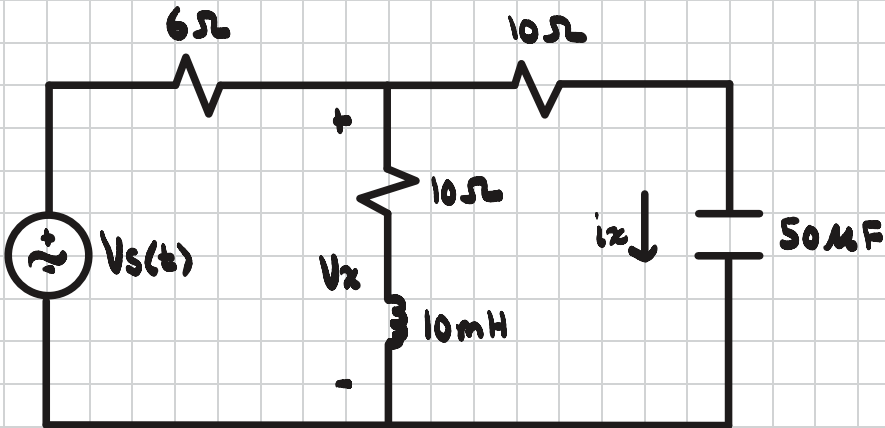
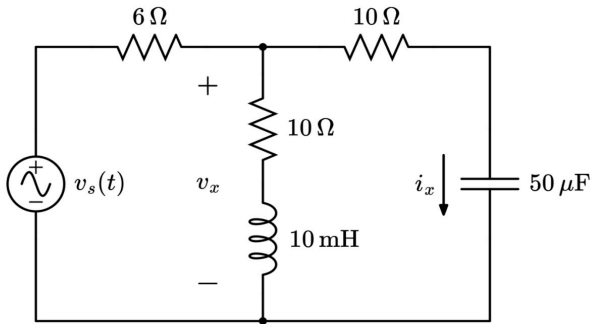


For the circuit shown below, find $v_x(t)$ and $i_x(t)$.

Suggested time limit: 10 minutes

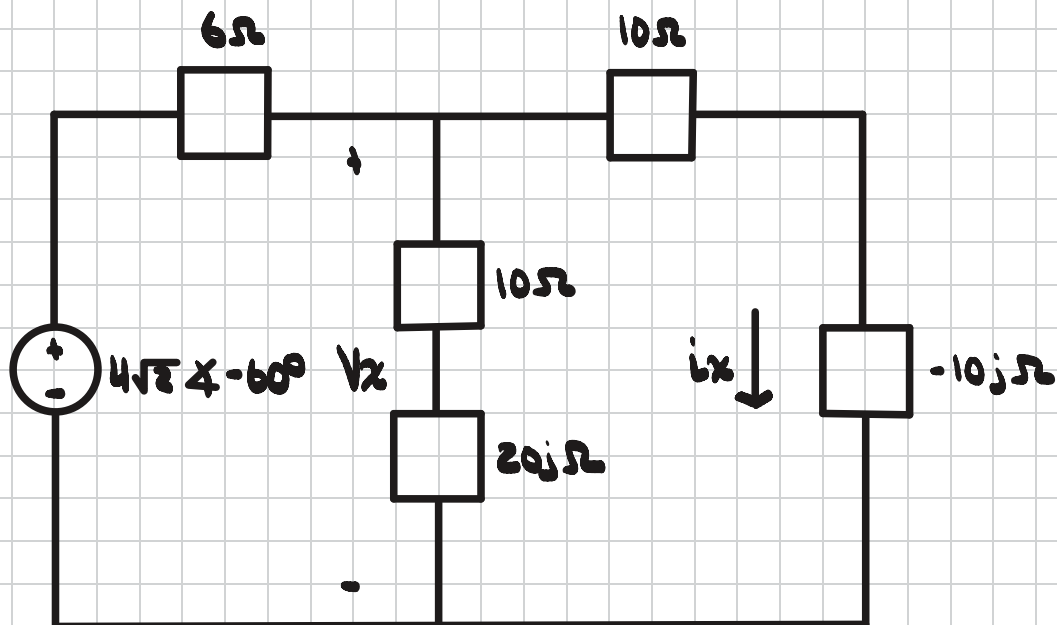
$$v_s(t) = 4\sqrt{2} \cos(2,000t - 60^\circ) \text{ V}$$



$$V_s = 4 \angle -60^\circ \text{ V}$$

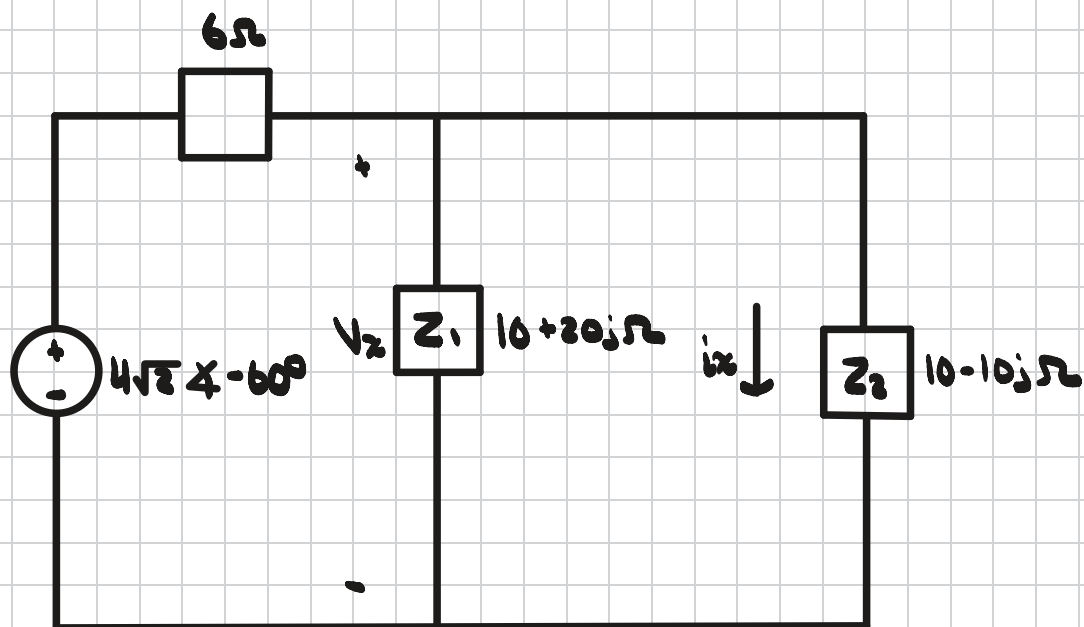
$$Z_L = j\omega L = j(2000)(10 \cdot 10^{-3}) = 20j \Omega$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(2000)(50 \cdot 10^{-6})} = -10j \Omega$$

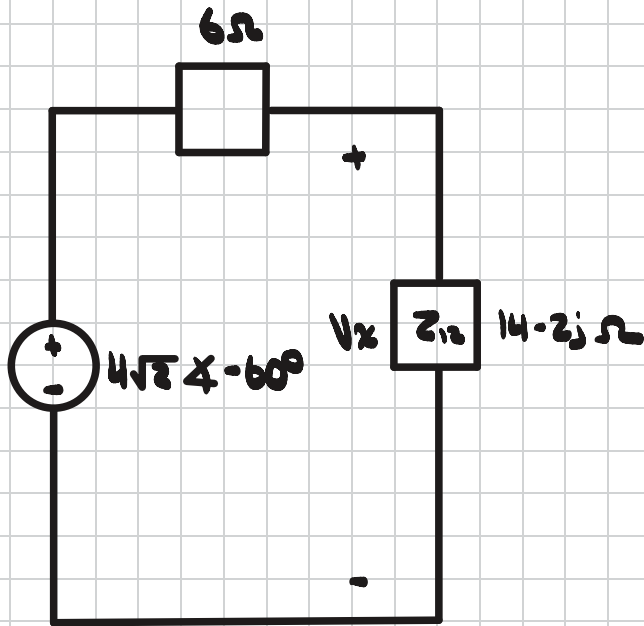


$$Z_1 = 10 + 20j\Omega$$

$$Z_2 = 10 - 10j\Omega$$



$$Z_1 || Z_2 = \left(\frac{1}{10+20j} + \frac{1}{10-10j} \right)^{-1} = 14-2j \Omega = Z_{12}$$



$$Z_{eq} = 6 + 14 - 2j = 20 - 2j \Omega$$

$$V_x = (4\sqrt{2} \angle -60^\circ) \left(\frac{14-2j}{20-2j} \right) = 3.98 \angle -62.42^\circ \text{ V}$$

$$i_x = \frac{3.98 \angle -62.42^\circ}{10-10j} = 0.28 \angle -17.42^\circ \text{ A}$$

$$V_x(t) = 3.98 \cos(2000t - 62.42^\circ) \text{ V}$$

$$i_x(t) = 0.28 \cos(2000t - 17.42^\circ) \text{ A}$$